

머신러닝 기초

Lecture 9: 벡터 미적분학

왜 벡터 미적분학을 배워야 할까?

- 머신 러닝 혹은 기계 학습
- 기계가 데이터를 보고 학습을 하는 법?
- 모델 혹은 함수가 데이터를 보고 에러를 계산한 뒤 에러를 줄이는 방향으로 학습
- 에러를 줄이는 방향?
 - 에러를 가장 빠르게 줄일 수 있는 방향 => Gradient!!
- 이번 장에서는 이 gradient를 계산하고 활용하는 법에 대해 배워 볼 예정

다루게 될 함수

- 다음과 같은 종류의 함수를 다룰 예정

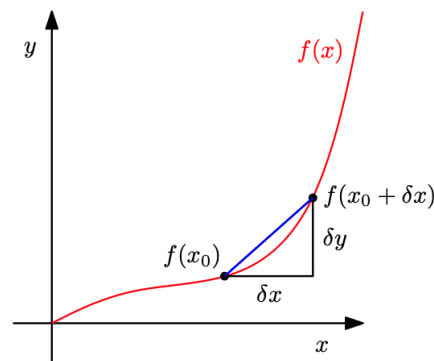
$$f : \mathbb{R}^D \rightarrow \mathbb{R}, \quad x \mapsto f(x)$$

평균 변화율

- 함수값의 평균 변화율

$$\frac{\delta f}{\delta x} := \frac{f(x + \delta x) - f(x)}{\delta x}$$

- 말 그대로 x 와 $x + \delta x$ 사이의 평균적인 함수값의 변화율을 계산한 값



미분

- 함수값의 미분
- 평균 변화율에서 δx 를 무한히 작은 값으로 보내서 x 의 근처에서의 변화율을 계산

$$\frac{df}{dx} := \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

테일러 급수

- 다루기 어려운 함수를 다루기 쉬운 다항식으로 근사하는 법
- 공학에서 가장 중요한 수학 기법 중 하나!!
- **n차 테일러 급수**의 정의

$$T_n(x) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

- 여기서 x_0 는 기준점이 되는 점

테일러 급수

- 테일러 급수

$$T_{\infty}(x) := \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

- 기준점 x_0 가 0인 경우를 Maclaurin 급수라고 부름

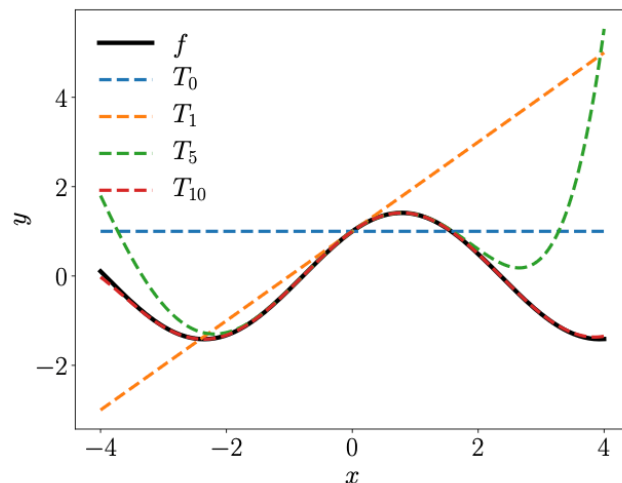
$$T_{\infty}(x) := \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

테일러 급수 예시

- $f(x) = \sin x + \cos x$
- $f(0) = \sin 0 + \cos 0 = 1$
- $f'(0) = \cos 0 - \sin 0 = 1$
- $f''(0) = -\sin 0 - \cos 0 = -1$
- $f^{(3)}(0) = -\cos 0 + \sin 0 = -1$

$$f^{(4)}(0) = \sin 0 + \cos 0 = 1$$

$$T_{\infty}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = 1 + x - \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 - \dots$$



미분 규칙

- 곱셈 규칙 $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$
- 분수 규칙 $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
- 덧셈 규칙 $(f(x) + g(x))' = f'(x) + g'(x)$
- 연쇄 법칙 $(g(f(x)))' = g'(f(x))f'(x)$

연쇄 법칙 예시

- $h(x) = (2x + 1)^4$
- $f(x) = 2x + 1$
- $g(f) = f^4$
- $f'(x) = 2, g'(f) = 4f^3$

$$h'(x) = g'(f)f'(x) = (4f^3) \cdot 2 = 4(2x + 1)^3 \cdot 2 = 8(2x + 1)^3$$

편미분

- $f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \mapsto f(x)$
- n차원 input을 받아서 1차원 output을 내놓는 함수에 대해서 input이 $x = (x_1, \dots, x_n)$ 일때 각 x_i 에 대한 미분을 **편미분**이라고 부름

$$\frac{\partial f}{\partial x_1} = \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(x_1, \dots, x_n)}{h}$$

$$\frac{\partial f}{\partial x_n} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{n-1}, x_n + h) - f(x_1, \dots, x_n)}{h}$$

편미분

- 이렇게 계산된 편미분들을 모은 것을 f 에 대한 **미분** 혹은 **gradient**라고 부름

$$\nabla_x f = \frac{df}{dx} = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right]$$

- 보통 사람들이 gradient를 계산하면 열 벡터로 많이 적음.
근데 이렇게 적으면 미분이 여러개 섞여 있는 경우 차원 맞추는 게 어려울 수 있음!!
- 항상 첫번째 차원을 f 차원을 두번째 차원을 x 차원을 넣자.

편미분 예시

- $f(x, y) = (x + 2y^3)^2$

$$\frac{\partial f(x, y)}{\partial x} = 2(x + 2y^3) \frac{\partial}{\partial x} (x + 2y^3) = 2(x + 2y^3)$$

$$\frac{\partial f(x, y)}{\partial y} = 2(x + 2y^3) \frac{\partial}{\partial y} (x + 2y^3) = 12(x + 2y^3)y^2$$

Gradient 예시

- $f(x_1, x_2) = x_1^2 x_2 + x_1 x_2^3$

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 2x_1 x_2 + x_2^3$$

$$\frac{\partial f(x_1, x_2)}{\partial x_2} = x_1^2 + 3x_1 x_2^2$$

$$\frac{df}{dx} = \left[\frac{\partial f(x_1, x_2)}{\partial x_1} \quad \frac{\partial f(x_1, x_2)}{\partial x_2} \right] = [2x_1 x_2 + x_2^3 \quad x_1^2 + 3x_1 x_2^2]$$

편미분의 기본 룰

- 곱셈 규칙

$$\frac{\partial}{\partial x}(f(x)g(x)) = \frac{\partial f}{\partial x}g(x) + f(x)\frac{\partial g}{\partial x}$$

- 합 규칙

$$\frac{\partial}{\partial x}(f(x) + g(x)) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}$$

- 연쇄 법칙 $\frac{\partial}{\partial x}(g \circ f)(x) = \frac{\partial}{\partial x}(g(f(x))) = \frac{\partial g}{\partial f} \frac{\partial f}{\partial x}$

연쇄 법칙 예시

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ 이고 x_1, x_2 는 t 의 함수, f 의 t 에 대한 미분은?

$$\frac{df}{dt} = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right] \begin{bmatrix} \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial t} \end{bmatrix}$$

$$\frac{df}{dt} = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right] \begin{bmatrix} \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial t} \end{bmatrix} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

연쇄 법칙 예시

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ 이고 x_1, x_2 는 s, t 의 함수, f 의 s, t 에 대한 미분은?

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial s} \quad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

$$\frac{df}{d(s, t)} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial (s, t)} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}$$

자코비안 행렬(Jacobian Matrix)

- $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\frac{df(x)}{dx} = \left[\frac{\partial f}{\partial x_1} \cdots \frac{\partial f}{\partial x_n} \right]$$

$$\begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \cdots & \frac{\partial f_1(x)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial x_1} & \cdots & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix}$$

- 이런 경우를 보통 특별히 **Jacobian matrix**라고 부르고

$$J = \nabla_x f$$

연쇄 법칙 예시

Example 5.10 (Chain Rule)

Consider the function $h : \mathbb{R} \rightarrow \mathbb{R}$, $h(t) = (f \circ g)(t)$ with

$$f : \mathbb{R}^2 \rightarrow \mathbb{R} \quad (5.69)$$

$$g : \mathbb{R} \rightarrow \mathbb{R}^2 \quad (5.70)$$

$$f(\mathbf{x}) = \exp(x_1 x_2^2), \quad (5.71)$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = g(t) = \begin{bmatrix} t \cos t \\ t \sin t \end{bmatrix} \quad (5.72)$$

and compute the gradient of h with respect to t . Since $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}^2$ we note that

$$\frac{\partial f}{\partial \mathbf{x}} \in \mathbb{R}^{1 \times 2}, \quad \frac{\partial g}{\partial t} \in \mathbb{R}^{2 \times 1}. \quad (5.73)$$

The desired gradient is computed by applying the chain rule:

$$\frac{dh}{dt} = \frac{\partial f}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial t} \end{bmatrix} \quad (5.74a)$$

$$= \begin{bmatrix} \exp(x_1 x_2^2) x_2^2 & 2 \exp(x_1 x_2^2) x_1 x_2 \end{bmatrix} \begin{bmatrix} \cos t - t \sin t \\ \sin t + t \cos t \end{bmatrix} \quad (5.74b)$$

$$= \exp(x_1 x_2^2) (x_2^2 (\cos t - t \sin t) + 2x_1 x_2 (\sin t + t \cos t)), \quad (5.74c)$$

where $x_1 = t \cos t$ and $x_2 = t \sin t$; see (5.72).

행렬 미분

Example 5.12 (Gradient of Vectors with Respect to Matrices)

Let us consider the following example, where

$$\mathbf{f} = \mathbf{A}\mathbf{x}, \quad \mathbf{f} \in \mathbb{R}^M, \quad \mathbf{A} \in \mathbb{R}^{M \times N}, \quad \mathbf{x} \in \mathbb{R}^N \quad (5.85)$$

and where we seek the gradient $d\mathbf{f}/d\mathbf{A}$. Let us start again by determining the dimension of the gradient as

$$\frac{d\mathbf{f}}{d\mathbf{A}} \in \mathbb{R}^{M \times (M \times N)}. \quad (5.86)$$

By definition, the gradient is the collection of the partial derivatives:

$$\frac{d\mathbf{f}}{d\mathbf{A}} = \begin{bmatrix} \frac{\partial f_1}{\partial \mathbf{A}} \\ \vdots \\ \frac{\partial f_M}{\partial \mathbf{A}} \end{bmatrix}, \quad \frac{\partial f_i}{\partial \mathbf{A}} \in \mathbb{R}^{1 \times (M \times N)}. \quad (5.87)$$

To compute the partial derivatives, it will be helpful to explicitly write out the matrix vector multiplication:

$$f_i = \sum_{j=1}^N A_{ij}x_j, \quad i = 1, \dots, M, \quad (5.88)$$

and the partial derivatives are then given as

$$\frac{\partial f_i}{\partial A_{iq}} = x_q. \quad (5.89)$$

This allows us to compute the partial derivatives of f_i with respect to a row of $\mathbf{A}$, which is given as

$$\frac{\partial f_i}{\partial \mathbf{A}_{i,:}} = \mathbf{x}^\top \in \mathbb{R}^{1 \times N}, \quad (5.90)$$

행렬 미분

$$\frac{\partial f_i}{\partial \mathbf{A}_{k \neq i, :}} = \mathbf{0}^\top \in \mathbb{R}^{1 \times 1 \times N} \quad (5.91)$$

where we have to pay attention to the correct dimensionality. Since f_i maps onto $\mathbb{R}$ and each row of $\mathbf{A}$ is of size $1 \times N$, we obtain a $1 \times 1 \times N$ -sized tensor as the partial derivative of f_i with respect to a row of $\mathbf{A}$.

We stack the partial derivatives (5.91) and get the desired gradient in (5.87) via

$$\frac{\partial f_i}{\partial \mathbf{A}} = \begin{bmatrix} \mathbf{0}^\top \\ \vdots \\ \mathbf{0}^\top \\ \mathbf{x}^\top \\ \mathbf{0}^\top \\ \vdots \\ \mathbf{0}^\top \end{bmatrix} \in \mathbb{R}^{1 \times (M \times N)}. \quad (5.92)$$

기억해두면 유용한 공식들

$$\frac{\partial}{\partial \mathbf{X}} f(\mathbf{X})^\top = \left(\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \right)^\top \quad (5.99)$$

$$\frac{\partial}{\partial \mathbf{X}} \text{tr}(f(\mathbf{X})) = \text{tr} \left(\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \right) \quad (5.100)$$

$$\frac{\partial}{\partial \mathbf{X}} \det(f(\mathbf{X})) = \det(f(\mathbf{X})) \text{tr} \left(f(\mathbf{X})^{-1} \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \right) \quad (5.101)$$

$$\frac{\partial}{\partial \mathbf{X}} f(\mathbf{X})^{-1} = -f(\mathbf{X})^{-1} \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} f(\mathbf{X})^{-1} \quad (5.102)$$

$$\frac{\partial \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{b}}{\partial \mathbf{X}} = -(\mathbf{X}^{-1})^\top \mathbf{a} \mathbf{b}^\top (\mathbf{X}^{-1})^\top \quad (5.103)$$

$$\frac{\partial \mathbf{x}^\top \mathbf{a}}{\partial \mathbf{x}} = \mathbf{a}^\top \quad (5.104)$$

$$\frac{\partial \mathbf{a}^\top \mathbf{x}}{\partial \mathbf{x}} = \mathbf{a}^\top \quad (5.105)$$

$$\frac{\partial \mathbf{a}^\top \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{b}^\top \quad (5.106)$$

$$\frac{\partial \mathbf{x}^\top \mathbf{B} \mathbf{x}}{\partial \mathbf{x}} = \mathbf{x}^\top (\mathbf{B} + \mathbf{B}^\top) \quad (5.107)$$

$$\frac{\partial}{\partial \mathbf{s}} (\mathbf{x} - \mathbf{A} \mathbf{s})^\top \mathbf{W} (\mathbf{x} - \mathbf{A} \mathbf{s}) = -2(\mathbf{x} - \mathbf{A} \mathbf{s})^\top \mathbf{W} \mathbf{A} \quad \text{for symmetric } \mathbf{W} \quad (5.108)$$

Q&A